

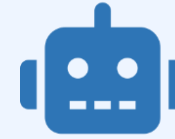
TCS349: Responsible and Explainable AI

Unit 3, Lecture 1

AI and Employment: Automation, Displacement & Changing Work

How AI automates tasks rather than whole jobs, which occupations face the most exposure, and how the structure of work itself is shifting as a result.

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Task Automation



Job Displacement



Occupational Structures



Labour Market Trends

1. Task Automation - What AI Actually Automates

AI typically automates specific tasks within a job, not the entire job itself - most occupations combine automatable and non-automatable tasks.

Tasks most exposed to automation tend to be routine, repetitive and rule-based, whether physical or cognitive.

Tasks that require complex judgement, physical dexterity in unpredictable settings, or interpersonal skill remain harder to automate.

Understanding automation at the task level, not the job level, gives a much more accurate picture of employment impact.

2. Which Jobs Are Most Exposed to Automation

Data entry, routine bookkeeping and basic customer support show high exposure, since their core tasks are repetitive and rule-based.

Manufacturing and warehouse roles involving repetitive physical tasks are highly exposed to robotic and AI-driven automation.

Roles requiring high emotional intelligence, creative judgement or complex physical adaptability - such as therapists, skilled trades and elder care - show lower exposure.

Exposure does not always mean elimination; it often means a significant change in what the role actually involves.

3. Job Displacement - How It Happens

Direct displacement occurs when a task or role is automated entirely and the position is no longer needed at the same scale.

Displacement is often gradual, happening through reduced hiring rather than sudden mass layoffs.

Displacement effects are uneven - concentrated in specific industries, regions and skill levels rather than spread evenly across the economy.

Workers in routine-task-heavy roles face significantly higher displacement risk than those in roles requiring adaptive, non-routine skills.

4. Case Example: Automation in Manufacturing & Logistics

Automated warehouses use robots for picking, sorting and moving inventory, substantially reducing the need for manual warehouse labour.

Self-checkout and automated inventory systems have reduced routine retail staffing needs in many large stores.

Automated quality-inspection systems using computer vision have reduced the need for manual visual inspection roles on production lines.

These changes have shifted labour demand toward equipment maintenance, systems oversight and exception handling instead of manual execution.

5. Case Example: Automation in Knowledge Work

AI-assisted drafting tools have reduced the time required for routine legal document review and first-draft contract generation.

Automated transcription and summarisation tools have reduced demand for purely transcription-focused roles in media and administration.

AI coding assistants have changed how much routine boilerplate code a software developer needs to write manually.

In each case, the technology reduced time spent on routine sub-tasks rather than eliminating the profession outright.

6. Changing Occupational Structures

Some occupations shrink in headcount as their routine tasks are automated, even while the underlying industry keeps growing.

Other occupations grow specifically because AI systems need to be built, managed, audited and maintained.

Job titles and role definitions shift over time, often blending previously separate skill sets - such as data literacy combined with domain expertise.

The overall structure of an industry's workforce can shift markedly even when total employment in that industry stays roughly stable.

7. New and Transformed Roles Emerging from AI

AI trainers and data annotators: roles created specifically to prepare and label the data AI systems learn from.

AI oversight and audit roles: positions responsible for monitoring model performance, fairness and compliance.

Prompt engineers and AI-integration specialists: roles focused on effectively directing AI tools within existing workflows.

Many traditional roles are being redefined to include AI-collaboration skills rather than being replaced outright.